

CMP-25 - Runtime Supervisor

Control runtime mode, service lifecycle, readiness, health, resource limits, deployment instances, degraded modes, quarantine, and restart policy.

Authority and state ownership

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 02, 27, 29, 34. Authoritative state: Current Runtime Mode, Service Instance Records, Desired and Observed State, health and readiness state, resource assignments, quarantine state. Explicitly non-owned: application domain state, work admission, recovery authority, release certification.

Critical invariant

Restart, locality, or process liveness never establishes readiness, correctness, authority, or recovery.

Logical interfaces

CMP-25-IF-01 / Command / Start Service

Command `Start Service` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-02 / Command / Stop Service

Command `Stop Service` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-03 / Command / Restart Service

Command `Restart Service` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-04 / Command / Enter Degraded Mode

Command `Enter Degraded Mode` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-05 / Command / Quarantine Service

Command `Quarantine Service` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-06 / Query / Get Service State

Query `Get Service State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-07 / Query / Get Health

Query `Get Health` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-08 / Query / Get Resource Use

Query `Get Resource Use` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-09 / Event / Service Ready

Event `Service Ready` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-10 / Event / Service Degraded

Event `Service Degraded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-11 / Event / Service Quarantined

Event `Service Quarantined` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-25-IF-12 / Event / Service Stopped

Event `Service Stopped` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Provisioned > Starting > Ready > Degraded > Quarantined > Recovering > Ready or Stopped. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: restart loop, health false positive, resource exhaustion, quarantine bypass, configuration drift. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-25-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-25-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-25-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-25-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-25-TST-005 - Critical invariant. Restart, locality, or process liveness never establishes readiness, correctness, authority, or recovery. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-25-TST-006 - Failure containment. Inject restart loop, health false positive, resource exhaustion, quarantine bypass, configuration drift. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-25-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-25-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-25-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-25-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.